ONLINE PYTHON

Learn Python

Install Module

main.py

```python
#SURFACE BLASTING DESIGN CALCULATOR
print("Enter numbers only")

BD = float(input("Enter Blasting Diameter (mm): "))
ρ = float(input("Enter Average In-Hole Density (kg/m**3): "))
a = float(input("Enter Spacing To Burden Ratio: "))
BH = float(input("Enter Bench Height (m): "))
rock = input("rock type (hard/soft): ").lower()
S = float(input("Spacing S (m): "))

#Stemming length calculation
if rock == "hard":
    SL = 20 * BD / 1000
    U_div = 3
else: #soft
    SL = 30 * BD / 1000
    U_div = 4

#Burden Calculation
if a == 1:
    pattern = "square"
    B = S / a
elif a == 1.15:
    pattern = "staggered"
    B = S / a
else:
    pattern = "any other"
    B = S / a

#Subdrill calculation
if rock == "hard":
    U = B / 3
else:
    U = B / 4

if rock == "hard":
    CL = BH - SL + U
else:
```

Ln: 40, Col: 9

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```
Enter numbers only
Enter Blasting Diameter (mm): 165
Enter Average In-Hole Density (kg/m**3): 840
Enter Spacing To Burden Ratio: 1
Enter Bench Height (m): 12
rock type (hard/soft): hard
Spacing S (m): 1.5
square
3.3
1.5
0.5
6121.288295365278


** Process exited - Return Code: 0 **
```

```python
    SL = 30 * BD / 1000
    U_div = 4

#Burden Calculation
if a == 1:
    pattern = "square"
    B = S / a
elif a == 1.15:
    pattern = "staggered"
    B = S / a
else:
    pattern = "any other"
    B = S / a

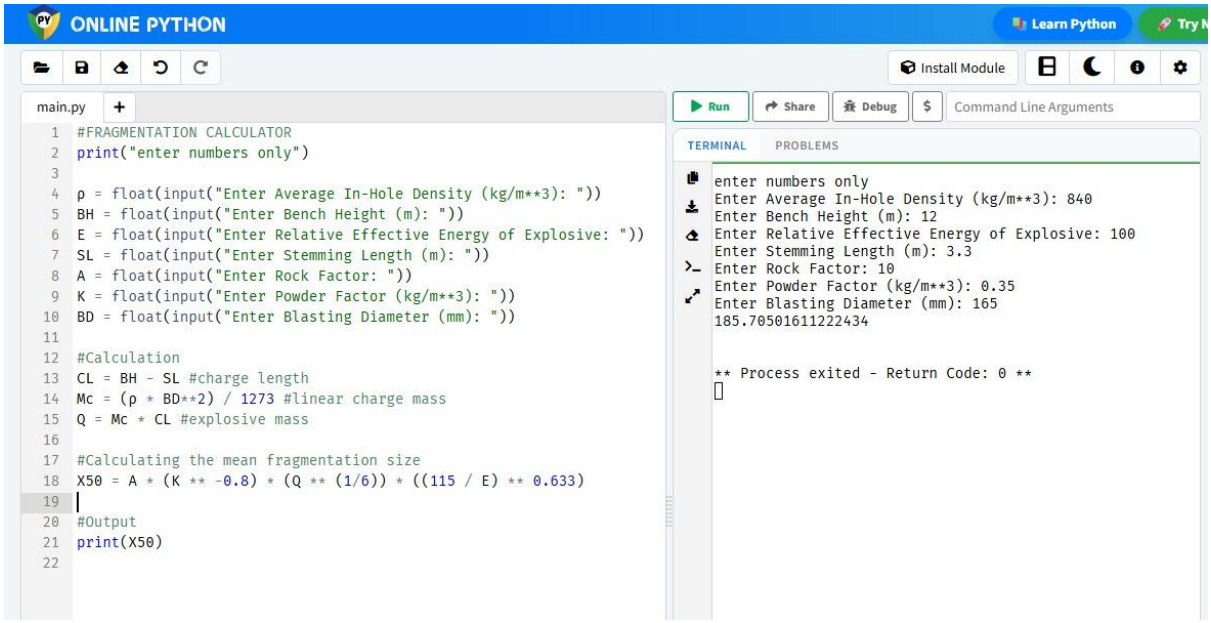
#Subdrill calculation
if rock == "hard":
    U = B / 3
else:
    U = B / 4

if rock == "hard":
    CL = BH - SL + U
else:
    CL = BH - SL - U
Mc = ( ρ * BD**2) / 1273 #linear charge mass
Kact = (CL * Mc) / (B * S * BH)

#Outputs
print(f"{pattern}")
print(f"{SL}")
print(f"{B}")
print(f"{U}")
print(f"{Kact}")
```

```
Enter Average In-Hole Density (kg/m**3): 840
Enter Spacing To Burden Ratio: 1
Enter Bench Height (m): 12
rock type (hard/soft): hard
Spacing S (m): 1.5
square
3.3
1.5
0.5
6121.288295365278


** Process exited - Return Code: 0 **
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#FRAGMENTATION CALCULATOR
print("enter numbers only")
ρ = float(input("Enter Average In-Hole Density (kg/m**3): "))
BH = float(input("Enter Bench Height (m): "))
E = float(input("Enter Relative Effective Energy of Explosive: "))
SL = float(input("Enter Stemming Length (m): "))
A = float(input("Enter Rock Factor: "))
K = float(input("Enter Powder Factor (kg/m**3): "))
BD = float(input("Enter Blasting Diameter (mm): "))
#Calculation
CL = BH - SL #charge length
Mc = (ρ * BD**2) / 1273 #linear charge mass
Q = Mc * CL #explosive mass
#Calculating the mean fragmentation size
X50 = A * (K ** -0.8) * (Q ** (1/6)) * ((115 / E) ** 0.633)
#Output
print(X50)
enter numbers only
Enter Average In-Hole Density (kg/m**3): 840
Enter Bench Height (m): 12
Enter Relative Effective Energy of Explosive: 100
Enter Stemming Length (m): 3.3
Enter Rock Factor: 10
Enter Powder Factor (kg/m**3): 0.35
Enter Blasting Diameter (mm): 165
185.70501611222434
** Process exited - Return Code: 0 **